

(a)

Heterogeneous Signals:

● Natural Language Prompt
(task description, instruction)

● Offline Designs & Associated Labels
● Response Design

You are a helpful optimization assistant tasked with generating a length-8 DNA sequence that maximizes binding affinity with a specific transcription factor (SIX6 REF R1). You are given several DNA sequences along with their corresponding binding affinities:

q {
|design-start| ['T', 'G', 'C', 'G', 'T', 'A', 'C', 'A'] |design-end| |label-start| +000.027 |label-end|
⋮
|design-start| ['A', 'C', 'G', 'T', 'A', 'C', 'G', 'T'] |design-end| |label-start| +000.018 |label-end|

Please propose a new DNA sequence that is different from the existing sequences in at least 1 position and has a higher binding affinity. Response:

o ← |design-start| ['G', 'C', 'G', 'T', 'A', 'C', 'G', 'T'] |design-end|



Unified Prompt – Response Corpus

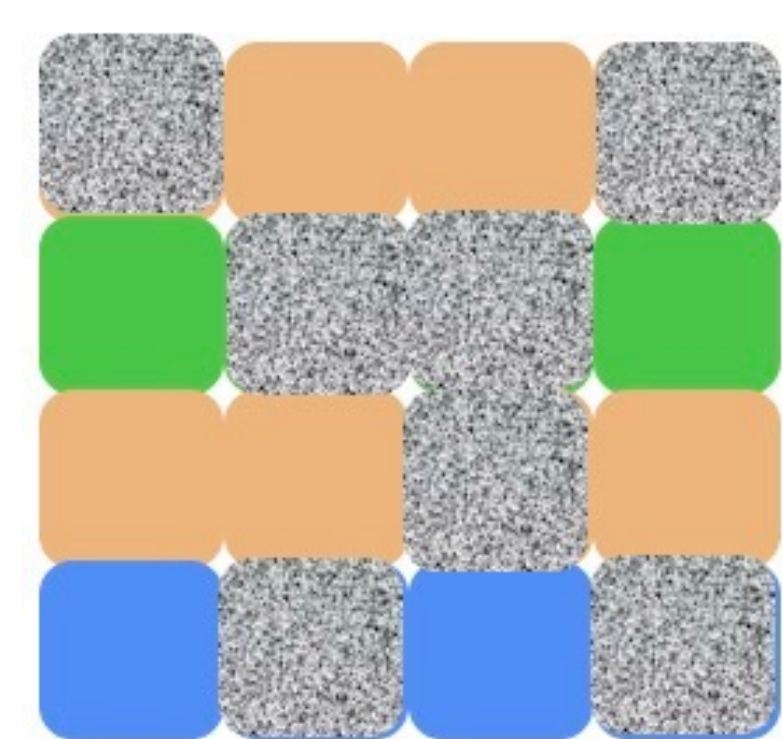
$$D_{text} = \{(q, o)\}$$

(b)

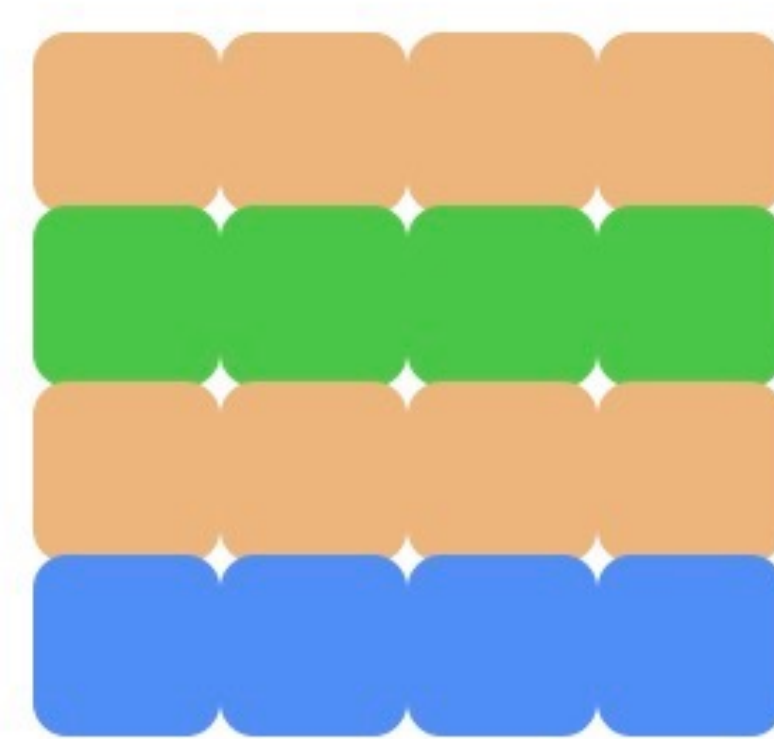
◆ $y(q)$: Max (Labels of Offline Designs)

◆ $y(o)$: Label of Response Design

Always have: ◆ $y(o) >$ ◆ $y(q)$



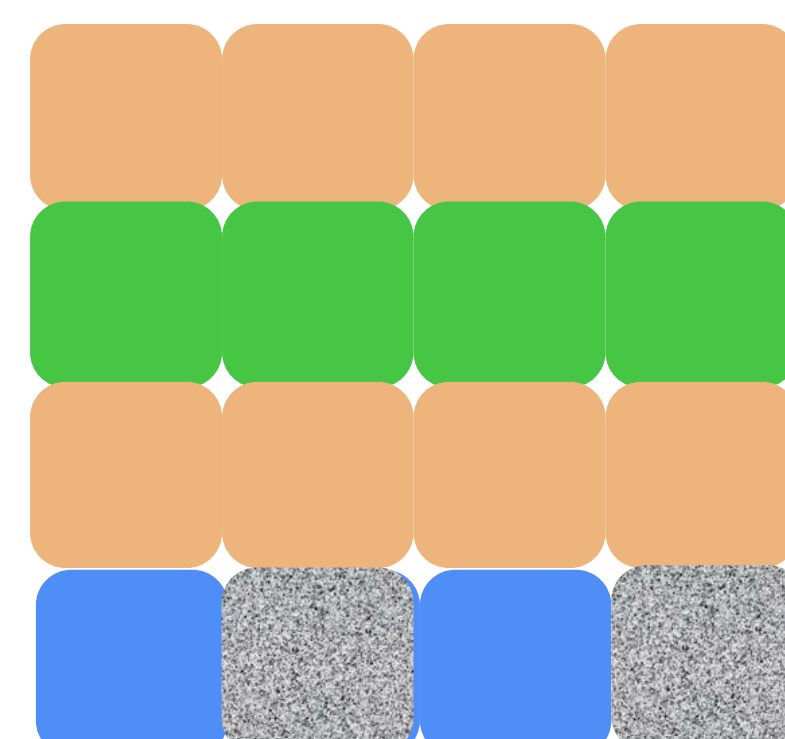
Denoise



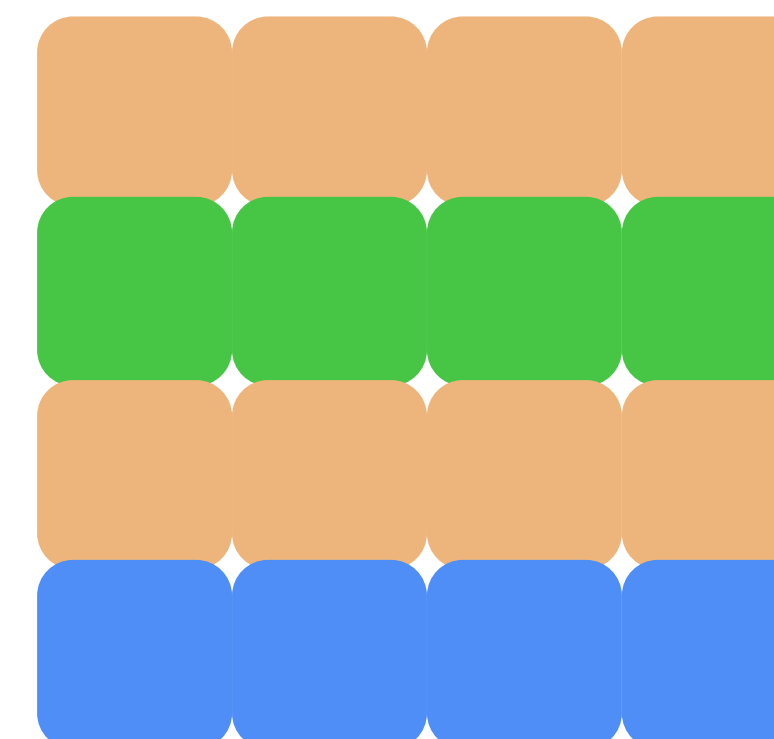
Domain Adaptation (DA)

(c)

Supervised Fine-Tuning (SFT)



Denoise



Post-Training

Reinforcement Learning (RL)

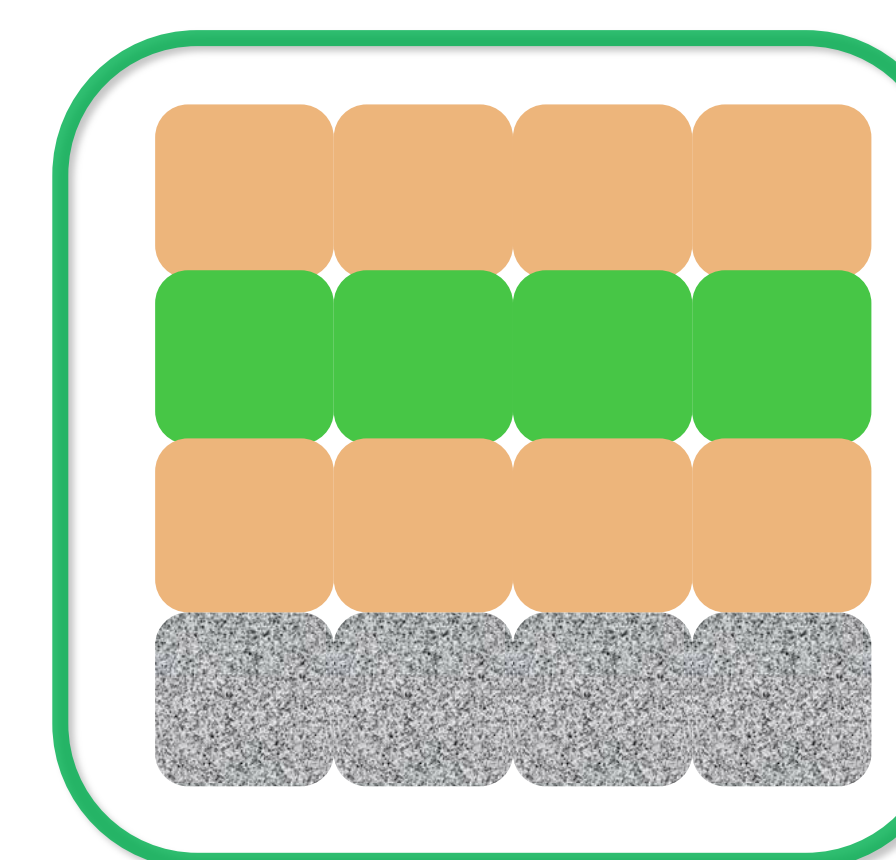


Reward Dataset

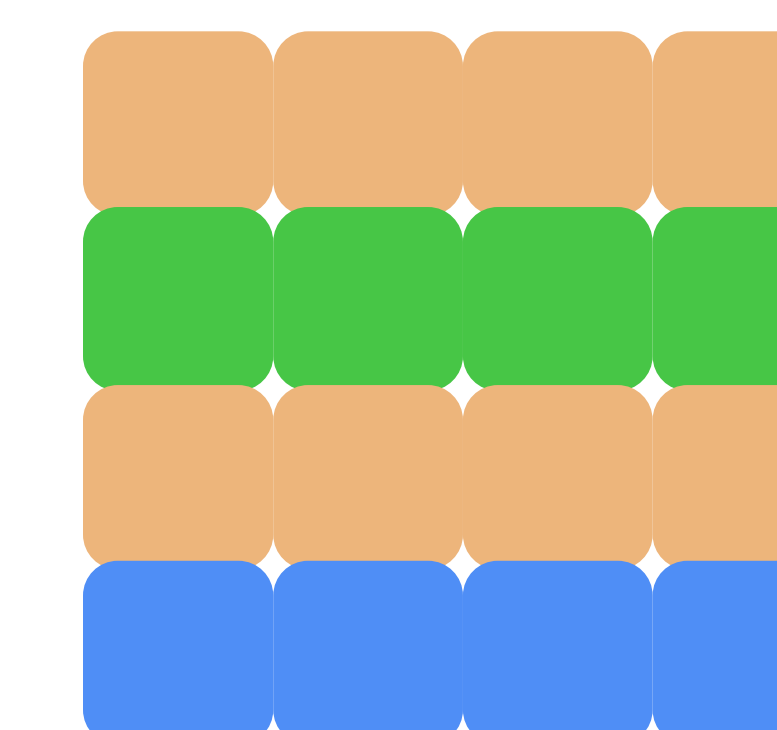
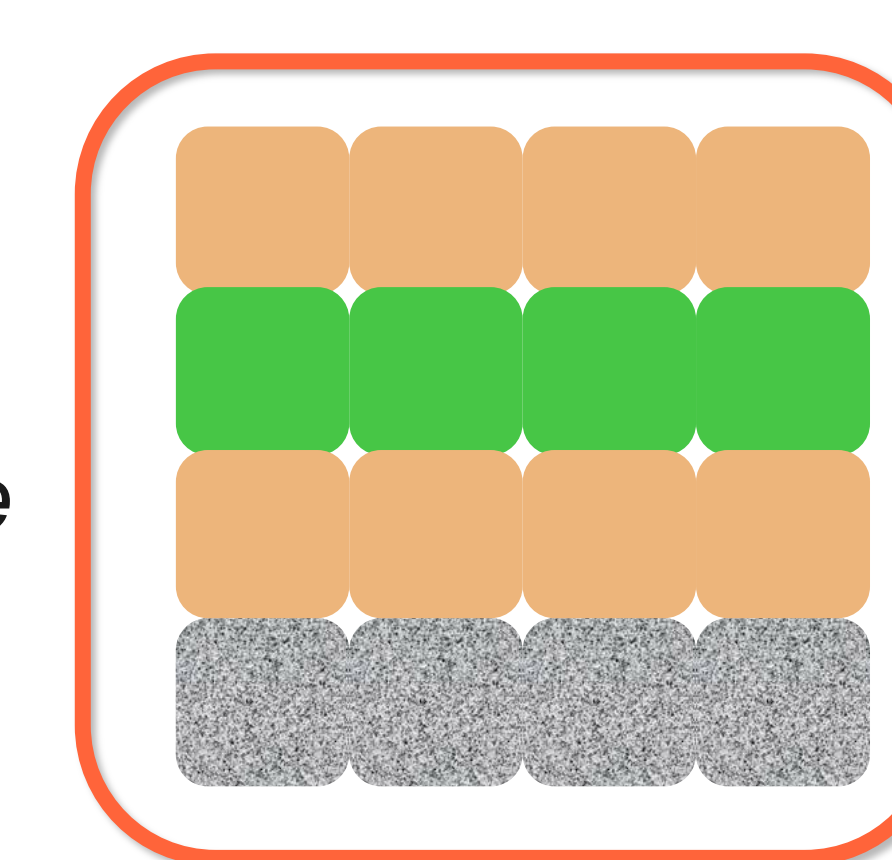
$$\star r(q, o) = \text{◆} - \text{◆}$$

★ < 0

★ > 0



Denoise



Tokens of Natural Language Prompt
 Tokens of Offline Designs & Associated Labels
 Tokens of Response Design
 Masked Tokens